
Practice test/50 Marks

I. Description of the Integrated situation

Ruhengeri Referral Hospital is both referral hospital and the only district hospital in Musanze District with 15 sectors, 68 cells and 432 villages it deserves a population of 406 557. It covers 16 health Centers;1 Prison dispensary and 13 health posts. The main mission is to improve the well-being of the population and promote the health of the population in Musanze however the Hospital serves part of neighboring districts BURERA, NYABIHU, GAKENKE and NYABIHU. The population from those districts is estimated to 386,078 and covers 18 health centres from those District.

Ruhengeri Referral Hospital was built in 1939 during the colonial period. It started its activities as a public clinic from 1939 to 1964.

In the 1980's Ruhengeri Referral Hospital was recognized as the national referral hospital. Since 1999, Ruhengeri Referral Hospital was ranked as a district hospital but continue to receive transfers from others neighboring district hospital (SHYIRA DH, NEMBA DH, GISENYI DH, BUTARO DH...).

Ruhengeri Referral Hospital is now recognized by government of Rwanda as Referral Hospital from 2013.

Ruhengeri Referral Hospital is requesting a C/C++ programmer to develop its healthcare system to manage patients' appointments using 3 linked lists. The thee linked lists and data for each are:

- ***PatientsLL (patient_id INTEGER, name STRING, dob STRING, gender STING)***
- ***DoctorsLL (doctor_id INTEGER, name STRING, specialization STRING)***
- ***Appointments (appointment_id INTEGER, patient_id INTEGER, doctor_id INTEGER, appointment_date STRING)***

As a software engineer, you are requested to develop this system for Ruhengeri Referral Hospital, and you are also requested to consider these:

- One `patient_id` references only one patient, one `doctor_id` references only one doctor and one `appointment_id` references only one appointment otherwise, notify the user that the given id already exists.
- The appointment is possible only if both ***patient_id*** and ***doctor_id*** exist in PatientsLL and DoctorsLL respectively. Otherwise, notify the user that one of them doesn't exist.
- The program will run on menus basis as follow:
 - 1: register patient
 - 2: register doctor
 - 3: register appointment
 - 4: display patients
 - 5: display doctors
 - 6: display appointments
 - 7: Exit

Please, refer to the following running sample of the application described above:

```
Menu:
1. Register a Patient
2. Register a Doctor
3. Register an appointment
4. Display Patients
5. Display Doctors
6. Display Appointments
7. Exit
Enter your choice: 1
PATIENT REGISTRATION
-----
ID: 1
NAME: Muhirwa
DoB: 12/12/2020
GENDER: Male
Menu:
1. Register a Patient
2. Register a Doctor
3. Register an appointment
4. Display Patients
5. Display Doctors
6. Display Appointments
7. Exit
Enter your choice: 2
DOCTOR REGISTRATION
-----
ID: 1
NAME: Massaissa
SPECIALIZATION: Hematology
Menu:
1. Register a Patient
2. Register a Doctor
3. Register an appointment
4. Display Patients
5. Display Doctors
6. Display Appointments
7. Exit
Enter your choice: 4
Patient ID: 1, Name: Muhirwa, DOB: 12/12/2020, Gender: Male
Menu:
1. Register a Patient
2. Register a Doctor
3. Register an appointment
4. Display Patients
```

```
5. Display Doctors
6. Display Appointments
7. Exit
Enter your choice: 5
Doctor ID: 1, Name: Massaissa, Specialization: Hematology
Menu:
1. Register a Patient
2. Register a Doctor
3. Register an appointment
4. Display Patients
5. Display Doctors
6. Display Appointments
7. Exit
Enter your choice: 3
APPOINTMENT REGISTRATION
-----
ID: 1
P_ID: 1
D_ID: 1
DATE: 20/2/2024
Menu:
1. Register a Patient
2. Register a Doctor
3. Register an appointment
4. Display Patients
5. Display Doctors
6. Display Appointments
7. Exit
Enter your choice: 6
Appointment ID: 1, Patient ID: 1, Doctor ID: 1, Date: 20/2/2024
Menu:
1. Register a Patient
2. Register a Doctor
3. Register an appointment
4. Display Patients
5. Display Doctors
6. Display Appointments
7. Exit
Enter your choice:
```
